

K-Means Clustering

Geometric Intuition, Mathematical Foundations, and Algorithmic Efficiency
for Unsupervised Learning.

Velix AI — AI. Simplified.

Core Concepts

Understanding the "Unsupervised" nature of finding patterns without labels.

What is Clustering?

The Unsupervised Goal

Clustering operates on **unlabeled data** to discover hidden structures. It groups data points such that objects within the same cluster exhibit high similarity.

Similarity vs. Variance

The core objective is to maximize intra-cluster similarity (closeness) while maximizing inter-cluster dissimilarity (separation between groups).

The Distance Rule



Euclidean Metric

K-Means relies on distance metrics to assign data samples to centroids. For an n -dimensional space, the distance is calculated as:

$$d(X_i, C_j) = \sqrt{\sum_{k=1}^n (x_{ik} - c_{jk})^2}$$

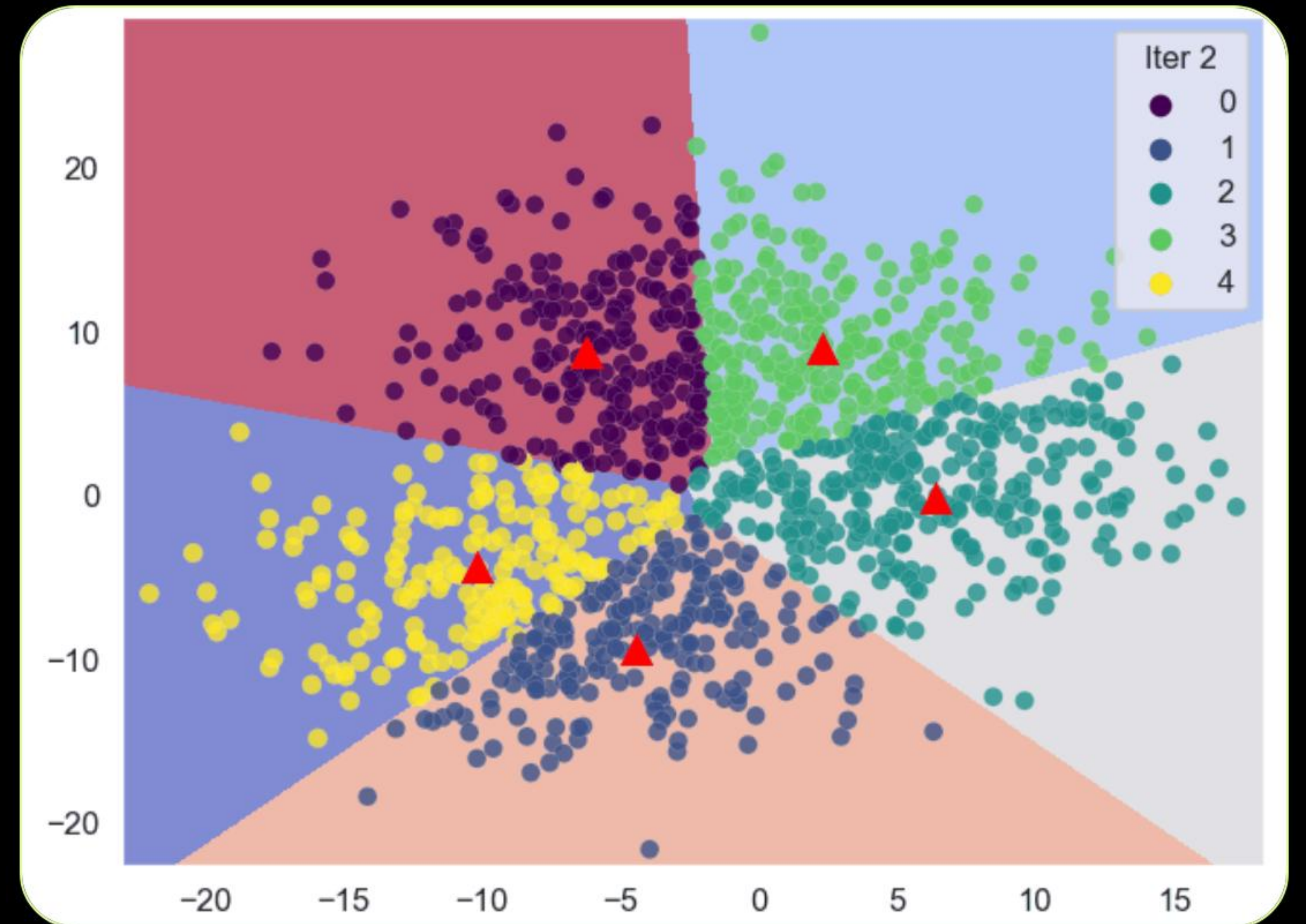
Initialization: K & Centroids

Step 1: Define Hyperparameter K

The algorithm cannot guess K. You must specify how many clusters to create based on domain knowledge or methods like the Elbow Plot.

Step 2: Random Placement

Initially, K random points (centroids) are placed in the feature space as "temporary" cluster centers.



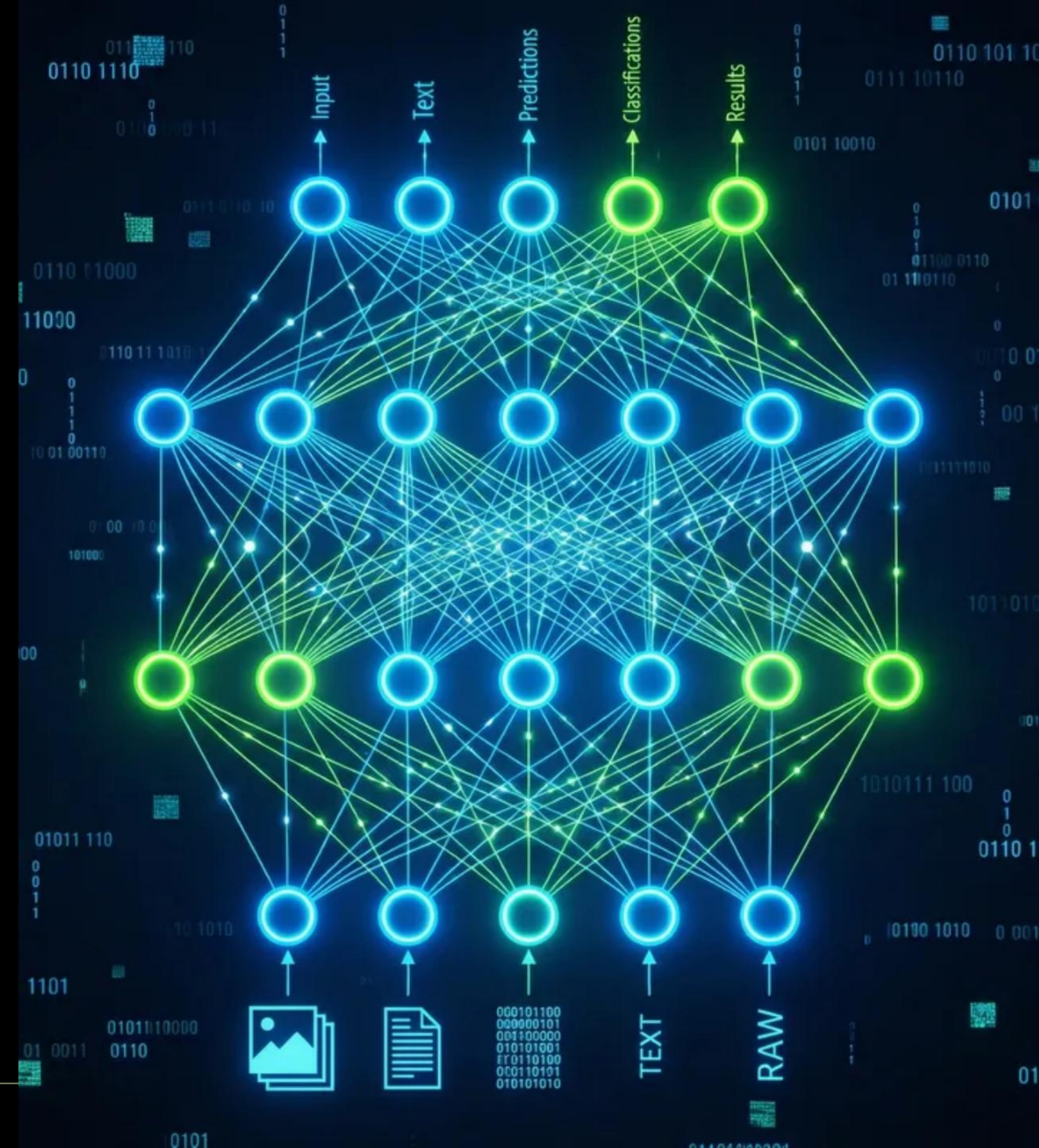
The Iterative Loop

Assignment & Update

Assignment: Every data point is assigned to its nearest centroid via Euclidean distance calculation.

Update: The centroid is re-calculated as the geometric mean of all points assigned to its cluster.

This deterministic loop repeats thousands of times until the system achieves mathematical stability.



Convergence Rules



Centroid Stability

The primary stopping condition is when the centroids no longer shift position after an update phase.



Max Iterations

To prevent infinite computation, a threshold for maximum iterations is set (usually 100-300 cycles).



Error Convergence

Termination also occurs when the reduction in variance (WCSS) falls below a pre-set tolerance level.

The Randomness Factor

Local Minima

Challenges of Initialization

K-Means is sensitive to the initial random placement of centroids. Bad luck in initialization can lead to "Local Minima," where the algorithm converges on a sub-optimal grouping.

Solutions like **K-Means++** use a spread-out initialization strategy to ensure more consistent and accurate global results.

Standard vs. K-Means++

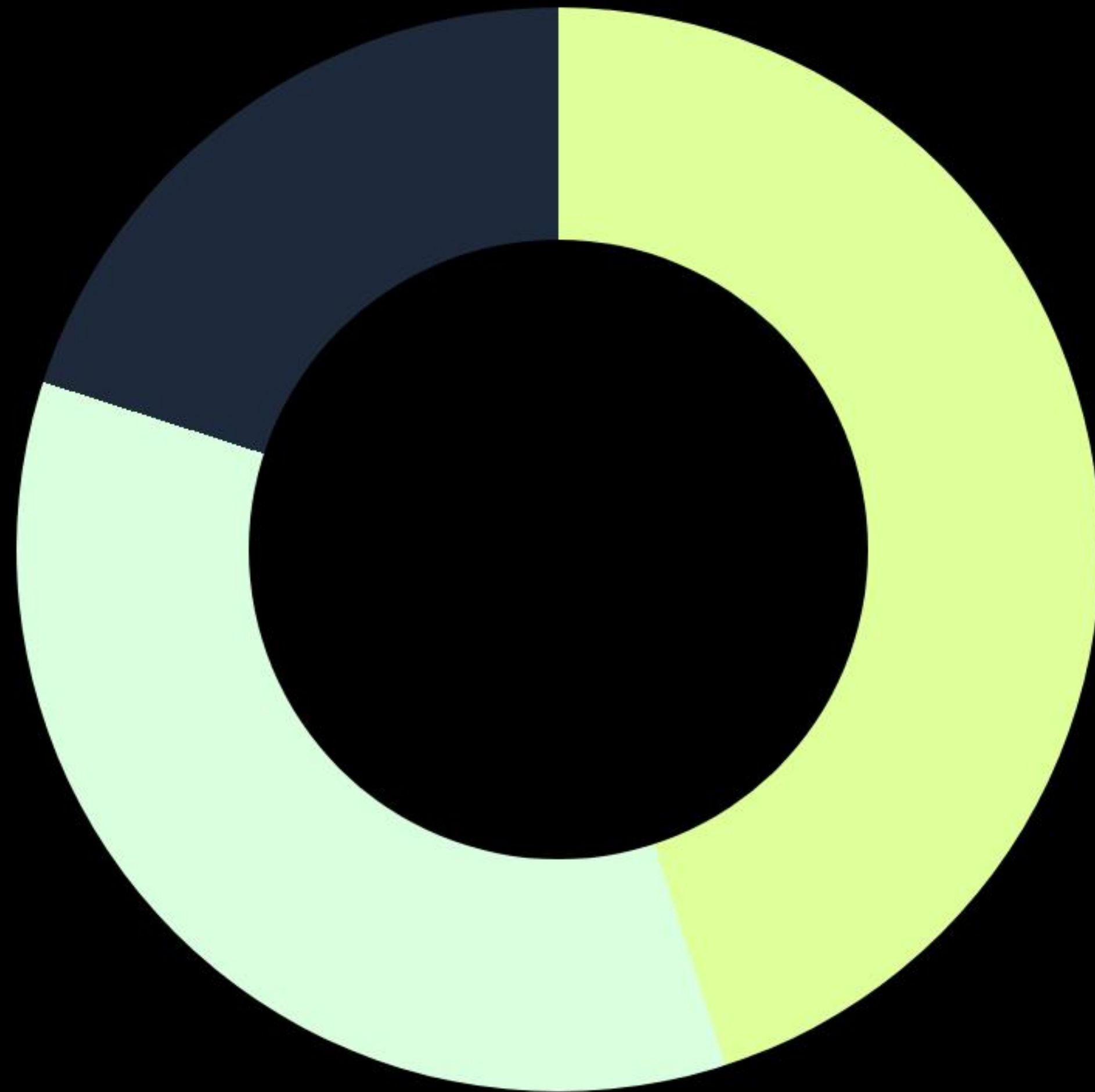
Feature	Standard K-Means	K-Means++
Initialization	Purely Random	Probability-weighted Spreading
Convergence Speed	Varies, can be slow	Generally Faster
Local Minima Risk	High	Significantly Reduced
Computational Cost	Low (Initial)	Higher (Initial)

Optimizing K: The Elbow Method



The Elbow Point is the specific index where the rate of drop in Within-Cluster Sum of Squares (WCSS) transitions from sharp to highly gradual.

Final Cluster Distribution



- Cluster A: Market Leaders (45%)
- Cluster B: Niche Specialists (35%)
- Cluster C: New Entrants (20%)

Questions & Next Steps

Thank you for attending this Velix AI Masterclass on Clustering Intuition. Feel free to ask about Silhouette Analysis or Mini-Batch K-Means.

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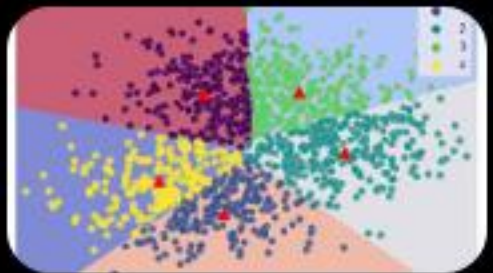
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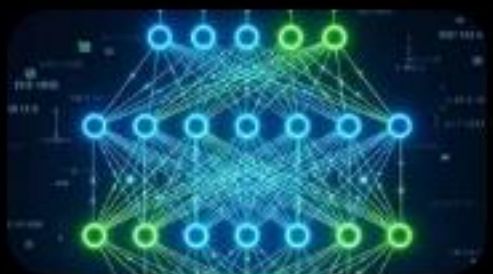
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